



Whether aristolochic acid directly drives hepatocarcinogenesis: comprehensive investigations from mutational signatures to animal models

Lu Yang¹ · Wanjun Lin¹ · Meina Shi¹ · Zifeng Huang¹ · Xuening Zhang¹ · Yanchao Yang¹ · Tong Chu¹ · Dayuan Zheng¹ · Dongfan Yang¹ · Wenzhe Ma¹

Received: 11 April 2025 / Accepted: 20 May 2025 / Published online: 30 May 2025
 © The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2025

Abstract

Aristolochic Acid (AA) is a class of naturally occurring compounds characterized by potent genotoxicity and carcinogenicity, predominantly found in plants of the *Aristolochia* genus. The 1993 Belgian weight-loss incident and subsequent research unequivocally established the association between AA and “Aristolochic Acid Nephropathy” (AAN) as well as Balkan Endemic Nephropathy (BEN), prompting global regulatory actions. AA undergoes nitroreduction to generate highly reactive intermediates, which form persistent DNA adducts, inducing A:T to T:A transversion mutations during DNA replication. In 2017, 78% (76/98) of hepatocellular carcinoma (HCC) samples collected from two Taiwanese hospitals exhibited the characteristic mutational signature COSMIC SBS22, sparking a global debate on whether AA contributes to liver carcinogenesis. This review systematically synthesizes the research progress on AA, encompassing its historical applications, metabolic mechanisms, genotoxic features, and its potential causal relationship with HCC. It highlights the necessity of employing multi-omics technologies to elucidate the regulatory networks of AA-metabolizing enzymes, developing non-invasive biomarkers, and rigorously validating carcinogenic mechanisms through animal models. Furthermore, strengthening the regulation of AA-containing herbal medicines and enhancing global exposure surveillance are imperative for mitigating the burden of liver cancer. This review provides a comprehensive perspective on the toxicological mechanisms and carcinogenic risks of AA, offering insights into precision prevention and intervention strategies.

Keywords Aristolochic acid · Hepatocellular carcinoma · Aristolochic acid nephropathy · Balkan endemic nephropathy · DNA adducts · Nucleotide excision repair

Abbreviations

AA	Aristolochic acid
BEN	Balkan endemic nephropathy
AAN	Aristolochic acid nephropathy
HCC	Hepatocellular carcinoma
NER	Nucleotide excision repair
CHN	Chinese herb nephropathy
FDA	Food and Drug Administration
IARC	International Agency for Research on Cancer
NR	Nitroreduction

NQO1	NAD(P)H:quinone oxidoreductase 1
CYPOR	Cytochrome P450 reductase
CYP1 A1/2	Cytochrome P450 1A1 and 1A2
XDH	Xanthine dehydrogenase
COX	Cyclooxygenase
AL-NOSO3	N-sulfonyloxyaristolactam
AL-NOAc	N-acetoxyaristolactam
AAIa	8-Hydroxyaristolochic acid I
GST	Glutathione S-transferase
GSH	Glutathione
AL-dA	7-(Deoxyadenosin-N ⁶ -yl) aristolactam
AL-dG	7-(Deoxyguanosin-N ² -yl) aristolactam
TC-NER	Transcription-coupled repair
GG-NER	Global genome repair
SBS22	Single base substitution 22
COSMIC	Catalog of somatic mutations in cancer
PVTTs	Portal vein tumor thromboses
HBV	Hepatitis B virus

✉ Wenzhe Ma
 wzma@must.edu.mo

¹ State Key Laboratory of Quality Research in Chinese Medicine & Faculty of Chinese Medicine, Macau University of Science and Technology, Macau 999078, China

HCV	Hepatitis C virus
AFB1	Aflatoxin B1
PH	Partial hepatectomy

Introduction

Aristolochic acid (AA), a class of naturally occurring nitro-phenanthrene compounds, is ubiquitously distributed within plants of the *Aristolochia* genus. Since the fifth century BCE, these plants have been integrated into traditional medicinal systems across ancient Greece, Rome, and various global regions due to their anti-inflammatory and analgesic properties (Scarborough 2011). However, with the advancement of modern toxicological research, the potent genotoxicity and carcinogenicity of AA have been progressively unveiled. The 1993 Belgian weight-loss incident first brought the nephrotoxicity of AA to international attention, and the pathological similarities between the resulting “Aristolochic Acid Nephropathy” (AAN) and Balkan Endemic Nephropathy (BEN) further underscored the transregional health hazards posed by AA (Danilovic et al. 1957; Gillerot et al. 2001). In 2017, the detection of the AA-specific

mutational signature COSMIC SBS22 in 78% of hepatocellular carcinoma (HCC) samples from Taiwan ignited intense academic debate regarding whether AA directly drives hepatocarcinogenesis (Ng et al. 2017).

The toxic mechanisms of AA are closely linked to its metabolic activation. Reactive intermediates generated through nitroreductive metabolism can covalently bind to DNA bases, forming persistent adducts that induce A:T → T:A transversion mutations. These mutations not only evade nucleotide excision repair (NER) but also exhibit distinct genomic signatures (Sidorenko et al. 2012). Nevertheless, the causal relationship between AA and HCC remains contentious: epidemiological studies reveal a significantly higher prevalence of SBS22 mutations among Asian HCC patients compared to other regions, yet direct exposure history is often lacking; animal studies yield conflicting results due to variations in administration routes and age-dependent responses, with adult models failing to replicate the carcinogenic effects observed in juvenile subjects (Fig. 1). Furthermore, confounding factors, such as co-exposure to HBV/HCV infections and aflatoxin, complicate causal inference.

This review systematically synthesizes the research progress on AA, encompassing its historical applications,

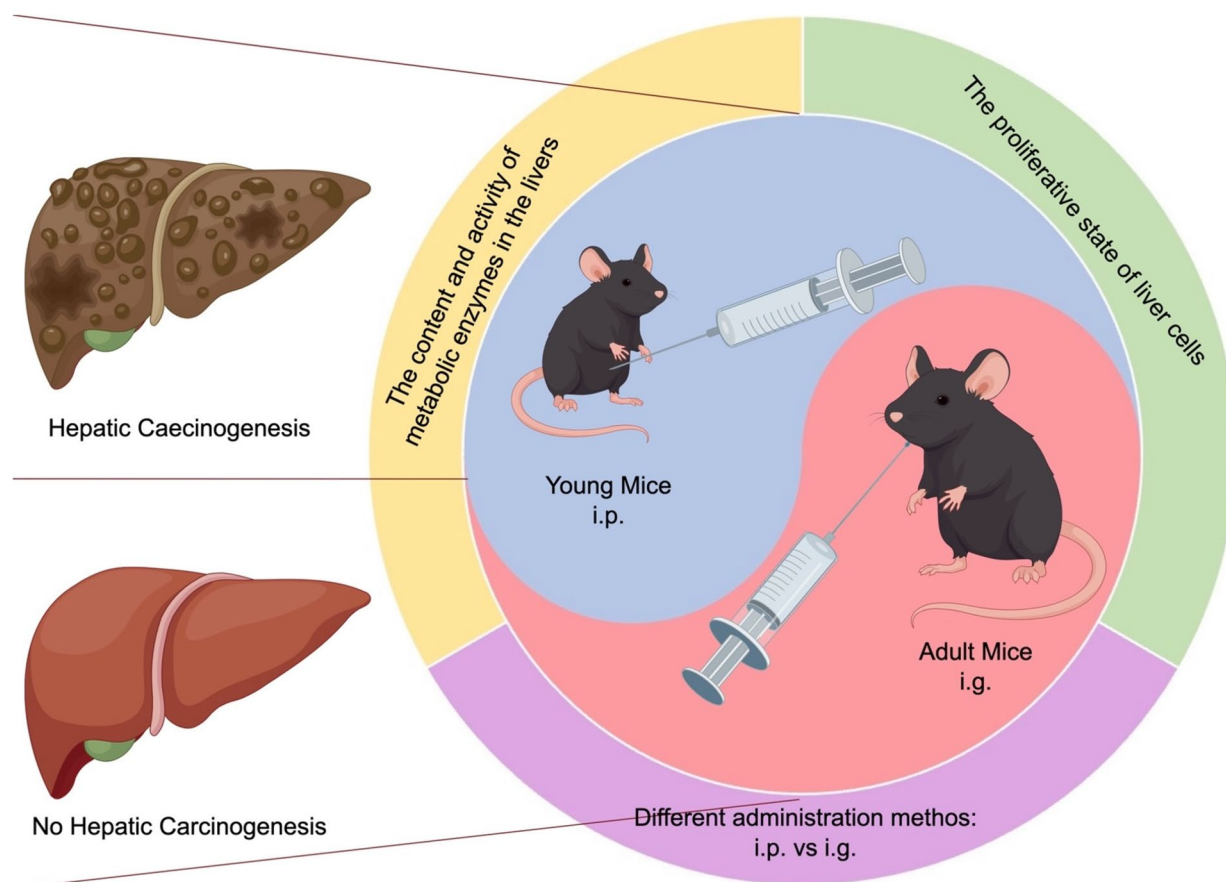


Fig. 1 Studies on AA and liver cancer produced contradictory results in young versus adult mice models exposed to AA. (By FigDraw)

metabolic pathways, genotoxic characteristics, and potential association with HCC. By integrating multi-disciplinary evidence, it aims to elucidate the mechanistic role of AA in hepatocarcinogenesis. Through the analysis of metabolic enzyme regulatory networks, the functional implications of mutational signatures, and the synergistic effects of multiple carcinogens, this review provides a scientific foundation for the development of precision prevention strategies. It also calls for enhanced regulation of AA-containing herbal medicines and global exposure surveillance to mitigate the burden of liver cancer.

Historical applications of AA

AA is a group of necrophilic carboxylic acids present in plants of *Aristolochia* species. The employment of plants harboring AA can be traced back to at least the fifth century BCE in ancient Greece and Rome to aid women in childbirth and treat gout (Scarborough 2011). *Aristolochia* species proliferate across every continent with the exception of Antarctica, and have played a prominent role in the traditional medical systems for upwards of two millennia, which was utilized as anti-inflammatory pharmaceuticals and analgesics (Tomlinson et al. 2020). It has been documented that diverse species of *Aristolochia* have been harnessed for medicinal purposes in numerous regions across the globe, predominantly in Asia, North America, India, Central and South America, Europe, and Africa (Heinrich et al. 2009).

In July 1953, Ganshirt H. first reported the isolation of a substance from plants of the *Aristolochiaceae* family and conducted a quantitative analysis. In the report, he named this substance “aristolochic acid” (Ganshirt 1953). In subsequent years, AA was increasingly recognized in human trials to exhibit anti-infectious, anti-inflammatory, and anti-neoplastic properties, and enhance the phagocytic capacity of immune cells (Moese 1963; Magliulo et al. 1996). Not until 1982 did toxicological experiments disclose the high carcinogenicity of AA. Subsequently, the West German health department mandated the withdrawal of all AA-containing drugs from the market (Hagemann et al. 1982). In 1964, China first documented a case where a patient suffered acute renal failure after ingesting *Aristolochiae Manshuriensis* Caulis, a Chinese medicine containing AA, to address postpartum lactation insufficiency (Songhan 1964). However, this case garnered no attention then.

During the 1920 s, an epidemic of interstitial nephritis emerged in the Danube River Basin, later known as the Balkan endemic nephropathy (BEN). BEN is a chronic tubulointerstitial nephropathy characterized by an insidious onset and gradual progression to end-stage renal disease (Danilovic et al. 1957). In 1956, Tanchev released the inaugural clinical account of BEN (Tanchev et al. 1956). Not

until 1967 did Kazantzis and Ivic first suggest that the accidental consumption of AA-containing wheat flour accounted for the BEN among farmers in the epidemic region (Ivić and Lovrić 1967). It was finally validated that chronic AA toxicity is the major pathogenic determinant of BEN (Stiborová et al. 2016). In subsequent investigations, researchers have discerned that while the majority of inhabitants in endemic regions may potentially be exposed to aristolochic acid, the prevalence of BEN remains below 10%. It is hypothesized that this phenomenon may be attributable to variations in the degree of exposure to AA among the local population, as well as genetic polymorphisms in enzymes responsible for the metabolism of AA within their systems.

In 1993, nine young Belgian women undergoing weight-loss treatment at a shared clinic suffered rapidly progressive interstitial renal fibrosis after ingesting traditional Chinese medicine capsules with AA. This incident propelled the nephrotoxicity of AA into the international limelight (Vanherweghem et al. 1993). Subsequently, this disease was named “Chinese Herb Nephropathy” (CHN). In 1994, upon further probing into the clinical traits of these female patients and the pathological attributes of nephroureterectomy specimens, researchers discovered an astounding resemblance between CHN and BEN, implying that AA shared etiological factor for both diseases (Cosyns et al. 1994). In the course of further investigation, CHN was eventually rechristened “Aristolochic Acid Nephropathy” (AAN) (Gillerot et al. 2001).

In 1999, the United Kingdom and Canada took the lead in proscribing the utilization of Chinese herbal remedies containing AA. Subsequently, nations and regions, including the United States, Germany, and Australia, implemented corresponding countermeasures.

In 2001, the U.S. Food and Drug Administration (FDA) promulgated a cautionary notice, indicating that the utilization of herbal products encompassing AA is perilous (Gold and Slone 2003). In 2002, The International Agency for Research on Cancer (IARC) categorized products encompassing *Aristolochia* species as carcinogenic agents for humans (IARC Working Group on the Evaluation of Carcinogenic Risks to Humans, International Agency for Research on Cancer, World Health Organization. Some traditional herbal medicines, some mycotoxins, naphthalene and styrene 2002). In 2003 and 2004, respectively, China revoked the medicinal standards for the AA-containing plants *Aristolochia manshuriensis*, *Radix Aristolochiae*, and *Aristolochia fangchi* (国家药品监督管理局.关于取消关木通药用标准的通知[EB, OL]. 2022; 国家食品药品监督管理局.关于加强广防己等6种药材及其制剂监督管理的通知[EB], 2023). From the 2005, 2010, and 2015 editions to the 2020 edition (Volume I) of the *Chinese Pharmacopoeia*, the inclusion of *Aristolochia*-genus herbs like *Aristolochia debilis*, *Herba Aristolochiae*, *Aristolochia manshuriensis*,

Aristolochia fangchi, and *Radix Aristolochiae* was gradually eliminated. Only the *Asarum*-genus herb *Asarum* was kept, with its medicinal part revised from the whole plant to roots and rhizomes (戴忠, 刘静, 郭日新等 2024).

Metabolic processes of AA

The metabolism of AA primarily encompasses two major pathways: bioactivation and detoxification (Fig. 2). Bioactivation generates highly reactive intermediates that bind to DNA, leading to carcinogenic effects. The detoxification pathway involves converting it into non-toxic metabolites through metabolic processes and reducing its toxicity via conjugation reactions.

Bioactivation

Nitroreduction

As with other nitroaromatic compounds, partial nitroreduction (NR) generates substances that are toxic and mutagenic (Penning et al. 2022). Among these, AL-NOH, formed through the four-electron NR of AA, serves as the direct precursor to the ultimate carcinogen nitrenium ion of AA. AL-NOH is a primary metabolite that is further metabolized into more reactive intermediates.

The enzymes involved in this process include:

NQO1 (NAD(P)H:quinone oxidoreductase 1): The most efficient enzyme in the bioactivation process of AA. It catalyzes the nitroreduction of AA-I/II to produce

N-hydroxyaristolactam (AL-NOH), primarily functioning in the liver and kidneys. It operates in the cytoplasm, transferring electrons via the FAD prosthetic group, with the active site residues Tyr128 and Tyr126 stabilizing and binding the substrate (Dong et al. 2023).

CYPOR (cytochrome P450 reductase): Works synergistically with NQO1 to participate in nitroreduction and significantly activates aristolochic acid in kidney microsomes.

CYP1 A1/2 (cytochrome P450 1 A1 and 1 A2): Relies on hydroxylated amino acids (Ser122/Thr124) to provide protons and catalyzes the reductive activation of AAI under hypoxic conditions, generating nitrenium ions (Levová et al. 2011).

Others: XDH (xanthine dehydrogenase) and COX (cyclooxygenase).

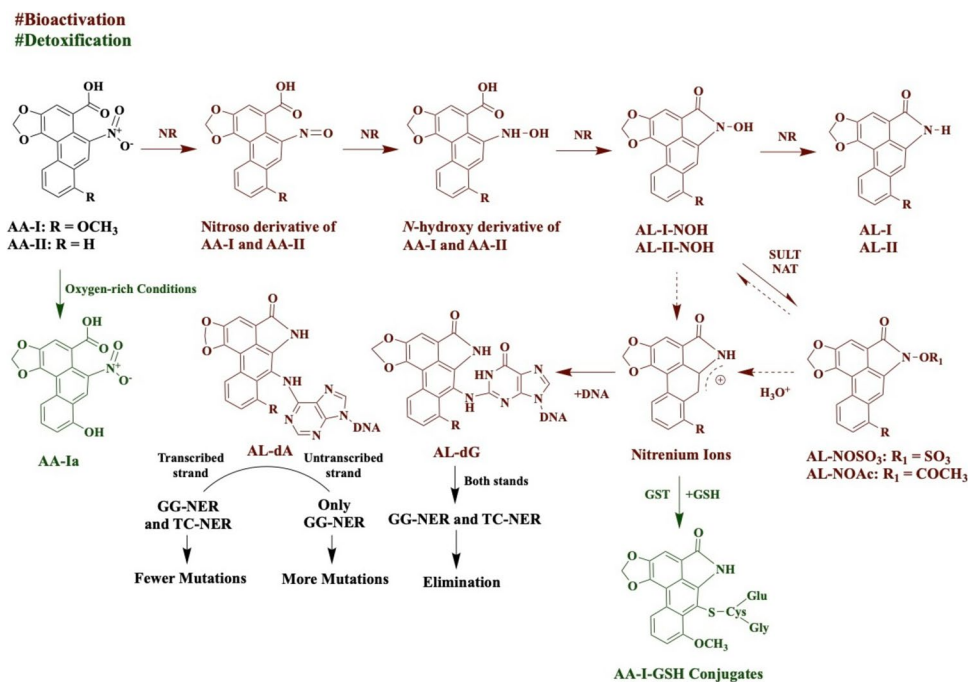
Sulfonation

SULT family (SULT1 A1, 1 A2, 1 A3, 1B1): Sulfonates AL-NOH in the liver to produce N-sulfonyloxyaristolactam (AL-NOSO₃). AL-NOSO₃ is relatively unstable and readily undergoes heterolytic cleavage to form cyclic nitrenium ions, which bind to DNA to form adducts (dA-AL and dG-AL), leading to gene mutations (Dubaisi et al. 2019).

Acetylation

NAT (N-acetyltransferases 1 and 2): Acetylates AL-NOH in the liver to produce N-acetoxyaristolactam (AL-NOAc). AL-NOAc also generates reactive intermediates through heterolytic cleavage, but its decomposition rate is slower, and

Fig. 2 The metabolic and mutagenic processes of AA in vivo



its efficiency in forming DNA adducts is lower than that of AL-NOSO3 (Dubaisi et al. 2019).

Detoxification

Conversion into non-toxic metabolites.

Under oxygen-rich conditions, CYP1 A1/2 (cytochrome P450 1 A1 and 1 A2) catalyzes the O-demethylation of AAI, producing AA1a (8-hydroxyaristolochic acid I). AA1a lacks DNA-binding capability and can be excreted through urine, significantly reducing renal toxicity (Rosenquist et al. 2010). Since CYP1 A2 is the predominant CYP isoform in the liver and CYP1 A1 is commonly found in renal tissue, both the kidneys and liver are capable of detoxifying AAI. AAII, lacking the 8-methoxy group, undergoes a different oxidative mechanism compared to AAI (Martínek et al. 2017).

Thiol conjugation

Glutathione S-transferase (GST): Catalyzes the conjugation of glutathione (GSH) with nitrenium ions to form AAI-GSH conjugates. Conjugation reactions deplete reactive intermediates, reducing the formation of DNA adducts and decreasing genotoxicity. Experimental evidence suggests that AAI-GSH conjugates are present at low concentrations in rat serum and may be excreted through bile (Zhang et al. 2020).

Since both AA-Ia and AL-DNA can be detected in the liver and kidneys of rodents exposed to AAI, it indicates that both organs are capable of detoxifying and activating AAI (Priestap et al. 2012). In other words, AAI can be metabolized through both oxidative and reductive pathways, whereas AAII relies almost exclusively on reductive activation, and the oxidative metabolic mechanism of AAII has not yet been fully elucidated. Evidence suggests that rat liver microsomes, which are rich in rodent CYP3 A, 2B, and 2E, metabolize AAII to produce AA-Ia (Martínek et al. 2017).

Genotoxicity of AA

The composition of AA is primarily characterized by the presence of AAI and AAII, with the difference between them being the presence or absence of an O-methoxy group at the C-8 position.

Formation of AA-DNA adducts

The primary activation pathway of AA involves nitroreduction, catalyzed by NAD(P)H:quinone oxidoreductase (NQO1). The nitro group is reduced to a hydroxylamine, yielding N-hydroxyaristolactam, which subsequently undergoes dehydration to form a highly reactive nitrenium ion.

This nitrenium ion then covalently binds to the amino groups of DNA purine bases (adenine N6 or guanine N2). AA forms covalent DNA adducts, respectively, 7-(deoxyguanosin-N²-yl) aristolactam (AL-dG) and 7-(deoxyadenosin-N⁶-yl) aristolactam (AL-dA), through metabolic activation (Fig. 2). AL-dA remains in rodent and human tissues with a half-life ranging from several months to years. In the context of the Belgian weight-loss incident, the DNA adducts AL-I-dA demonstrated significant long-term persistence in the renal tissues of AAN patients, lasting up to 2 decades, suggesting that these types of adducts may be resistant to repair. In contrast, AL-dG is relatively unstable and can be readily eliminated, disappearing from tissues within a short period of time. In human tissues, the predominant DNA adduct formed is AL-I-dA, while AL-I-dG and AL-II-dA constitute the minor DNA adducts (Chen et al. 2012). Consequently, AL-I-dA is frequently regarded as molecular biomarkers indicative of AA exposure.

DNA damage repair process induced by AA-DNA adducts

In humans, nucleotide excision repair (NER) serves as the primary pathway responsible for the removal of bulky DNA lesions. This process encompasses two sub-pathways: transcription-coupled repair (TC-NER) and global-genome repair (GG-NER). Upon the identification of DNA damage, these two sub-pathways collaborate to facilitate the completion of the repair process (Gillet and Schärer 2006). TC-NER specifically targets lesions on the transcribed strands of actively expressed genes and is initiated when RNA polymerase stalls at the site of damage (Hanawalt and Spivak 2008). In contrast, GG-NER surveils the entire genome and is initiated by the damage sensor XPC-RAD23B, which indirectly recognizes lesions by binding to the undamaged strand opposite the site of damage (Sugasawa et al. 1998).

Upon the formation of AA-DNA adducts, the human body initiates the NER mechanism. In this process, the AL-dG adducts on both the transcribed and non-transcribed strands can be recognized by the XPC protein and subsequently removed by GG-NER. During transcription, when RNA polymerase encounters and stalls at an AL-dG adduct on the transcribed strand, the lesion can also be cleared by the activated TC-NER pathway. As a result, AL-dG adducts are eventually eliminated from the tissues of both rodents and humans. However, in the case of AL-dA, the lesion site exhibits less distortion, stable stacking, and reduced dynamics, which hinder the insertion of the principal recognition motif of XPC, a β -hairpin, into the double helix. This impedes the binding of the adduct to XPC-RAD23B, making the recognition and repair process less efficient (Kathuria et al. 2015). This results in AL-dA not being cleared by GG-NER.

but instead relying on TC-NER to remove lesions on the transcribed strand. Consequently, AL-dA on the non-transcribed strand persists in the target organs for an extended period (Fig. 2).

DNA replication errors

Research indicates that AA preferentially binds to the context sequence of 5'-pyrimidine-A-purine-3' (such as 5'-CTG-3' or 5'-CAG-3'). Both dG-AL and dA-AL adducts can induce the incorporation of dAMP during DNA replication, ultimately leading to G:C to T:A and A:T to T:A transversions, respectively (Sidorenko et al. 2012). The primary adducts remaining in the target organs are AL-dA on the non-transcribed strand, which are left behind during the DNA damage repair process. Consequently, the mutations induced by AA predominantly manifest as A:T to T:A transversions within the 5'-pyrimidine-A-purine-3' context sequence and are almost entirely localized to the non-transcribed strand. The mutational signature Single Base Substitution 22 (SBS22) in the Catalog of Somatic Mutations in Cancer (COSMIC) database has been confirmed to be associated with aristolochic acid exposure. In the COSMIC SBS22 dataset, the A:T to T:A signature substitutions exhibit a pronounced predilection for the 5'-PyrAPur-3' sequence context, with the predominant substitutions being C[T to A]G and C[T to A]A (Alexandrov et al. 2020). In addition, recently, a comprehensive whole-genome analysis of 494 hepatocellular carcinoma cases revealed two additional mutational signatures associated with aristolochic acid exposure, apart from SBS22. These signatures include TA-to-NT, TC-to-AA, TG-to-AN, and TT-to-AA mutations, as well as 1 bp and 2 bp deletions in short repeats (Chen et al. 2024).

COSMIC SBS22 is particularly prevalent in the TP53 gene, with the induced mutations predominantly clustering in the TP53 mutation hotspot regions spanning exons 5–8 (Moriya et al. 2011). These mutations result in missense or nonsense changes that disrupt the function of the DNA-binding domain of the P53 protein, impairing cell cycle regulation and DNA repair capabilities, thereby promoting tumorigenesis. For instance, A:T to T:A transversions at codon 291 and codon 305 lead to the premature generation of stop codons (Nedelko et al. 2009). Existing studies have proposed that this TP53 mutation signature associated with AA could serve as a biomarker for AA exposure history. However, in practical applications, it should be evaluated comprehensively alongside the detection results of specific AA-DNA adducts and the individual's AA exposure history (Moriya et al. 2011).

The controversy over whether AA directly drives hepatocarcinogenesis

Hepatocellular carcinoma (HCC), the predominant form of liver cancer, ranks as the sixth most frequently diagnosed cancer globally and stands as the third most common cause of cancer-related deaths worldwide (Llovet et al. 2024).

COSMIC SBS22

Accumulated evidence demonstrates that exposure to AA induces the formation of AA-DNA adducts in hepatocytes across both human and rodent models (Lord et al. 2004; Dong et al. 2006). Furthermore, in a study investigating the distribution of DNA adducts across various organs following AA exposure, it was observed that 2 days post-administration, the concentration of DNA adducts in the rat liver was second only to that in the renal cortex and renal medulla (Liu et al. 2019). As early as 2012, in a study utilizing exome sequencing to identify somatic mutations in ten HBV-positive individuals with HCC with portal vein tumor thromboses (PVTs), intrahepatic metastases, T:A to A:T transversion was significantly enriched in four affected individuals (Huang et al. 2012). This finding markedly diverges from the mutation patterns previously reported in the other studies involving exome sequencing of hepatocellular carcinoma, prompting the authors to hypothesize a potential association with exposure to aristolochic acid or vinyl chloride. The following year, a subsequent study screened genomic or exome sequence data from 93 HBV-positive HCC cases, identifying a strong AA mutation signature in 11 of these cases. However, in the absence of definitive evidence indicating a history of AA exposure in these patients, the authors could only speculate that these 11 individuals might have had prior contact with AA (Poon et al. 2013). It was not until 2017 that exome sequencing of 98 HCC tissues from two hospitals in Taiwan revealed that 78% of the HCC cases exhibited the distinctive mutational signature COSMIC SBS22, indicative of AA exposure. Furthermore, sequencing of 1,400 HCC cases from diverse geographical regions demonstrated that the prevalence of COSMIC SBS22 varied significantly, with proportions of 47%, 29%, 13%, 2.7%, 4.8%, and 1.7% observed in cases from China, Southeast Asia, Korea, Japan, North America, and Europe, respectively (Ng et al. 2017). It is noteworthy that this study did not investigate the exposure background of the patients, and the tumor tissues were obtained from non-consecutive patients. The selection of HCC samples was based solely on the availability of sufficient DNA, rather than being driven by any prior suspicion of AA exposure.

Thus emerged the first article revealing the association between AA and the occurrence of HCC, sparking international controversy. This report suggests that liver cells are not only capable of bioactivating AA but may also serve as a target organ for AA. Subsequently, another analysis of public HCC datasets revealed that the prevalence of COSMIC SBS22 among patients treated in China, Singapore, and Korea was 26% (133/510), 44% (4/9), and 7% (16/231), respectively (Lu et al. 2020). In contrast, the observed prevalence of COSMIC SBS22 in HCC patients from Japan and France was less than 1%. Similarly, among HCC patients in the United States, the detection rate of COSMIC SBS22 was higher in Asian patients (24/160, 15%) compared to non-Asian patients (5/204, 2.5%). These findings further underscore the widespread presence of the unique mutational signature of AA among Asian HCC patients.

However, the sequencing results alone are insufficient to conclusively establish aristolochic acid as a causative agent of liver cancer. This is because these studies did not definitively ascertain the specific AA exposure history of the patients, nor did they measure the presence of AA-DNA adducts in the target organs. Additionally, it is important to note that AA is not the only chemical capable of inducing A:T to T:A transversions (Table 1). In China and Africa, the HBV genome exhibits a double A:T to T:A transversion at nucleotide positions 1762 and 1764 (1762 T/1764 A) (Tong 2005). DMBA induces a mutation pattern in codon 61 of the HRAS gene in mice that is strikingly similar to that caused by AA, with over 60% of the mutations being identical. The mutation hotspot also occurs at 5'-CAG/CTG-3', and DMBA is closely associated with the development of liver cancer (Nassar et al. 2015). In the sequence contexts of 5'-CTT-3' and 5'-CTG-3', the probability of such transversion mutations occurring in DEN-induced liver cancer is 37.3% (Connor et al. 2018). Furthermore, substances capable of inducing A:T to T:A transversion mutations include vinyl chloride and its reactive metabolite chloroethylene

oxide, both associated with liver cancer; tobacco carcinogens such as 4-aminobiphenyl and 1,3-butadiene; aflatoxin, a known liver carcinogen; chemotherapeutic agents melphalan and chlorambucil; as well as alcohol and its metabolite acetaldehyde (Hollstein et al. 1994; Garaycochea et al. 2018).

To investigate the reliability of the aforementioned sequencing results, researchers utilized liver cancer samples from Asian and Caucasian populations in the TCGA dataset to validate the frequency of AA mutation signatures. The results indicated that Asians exhibited a higher prevalence of AA mutation signatures among HCC patients, but the rate was only 10.2% (19/187), significantly lower than the findings reported in the previous studies (Ji et al. 2018). In 2022, another study randomly selected 1256 liver cancer samples from three medical centers, and only 5.10% of the samples were found to contain AAI-DNA adducts. Whole-genome sequencing results revealed that the frequency of COSMIC SBS22 mutations was merely 8.41% (9/107). This relatively low frequency may be attributed to the ban on the sale of AA-containing traditional Chinese medicines in China after 2003. Additionally, the sample size of this study was relatively small, potentially limiting its ability to represent the overall situation in China (Chen et al. 2022).

Clinical trials

Following the discovery of a high prevalence of COSMIC SBS22 in HCC patients, a retrospective, population-based cohort study conducted in 2018 on patients aged over 18 years diagnosed with HBV infection revealed that the intake of AA in HBV-infected patients dose-dependently increased their risk of developing HCC (Chen et al. 2018). Subsequently, two articles have reported that the intake of AA also increases the risk of HCC in patients with HCV infection and diabetes mellitus (Chen et al. 2019; 2022). However, the development of HCC is strongly linked to several major risk factors, including chronic infection

Table 1 Chemicals capable of inducing T:A to A:T transversions

Chemical	% of A:T to T:A	Strand bias	Trinucleotide	HCC
AA (Sidorenko et al. 2012)	> 50%	Non-transcribed	CAG/CTG	Unknown
DMBA (Nassar et al. 2015)	> 60%	Unknown	CAG/CTG	Yes
DEN(Connor et al. 2018)	37.3%	Unknown	CAT/CTT	Yes
Vinyl chloride (Hollstein et al. 1994)	Unknown	Unknown	Unknown	Yes
Ethylene oxide (Hollstein et al. 1994)	Unknown	Unknown	Unknown	Yes
4-Aminobiphenyl (Manjanatha et al. 1996)	Unknown	Unknown	Unknown	Yes
1,3-Butadiene (Ma et al. 2000)	Unknown	Unknown	Unknown	No
Aflatoxin(Huang et al. 2017)	5%	Non-transcribed	Unknown	Yes
Melphalan and chlorambucil (Wang et al. 1990)	Unknown	Unknown	Unknown	No
Alcohol and aldehyde (Garaycochea et al. 2018)	< 15%	Unknown	Unknown	Yes

with hepatitis B virus (HBV) or hepatitis C virus (HCV), excessive alcohol consumption, and dietary exposure to aflatoxin B1 (AFB1)-contaminated food. In the aforementioned experiments, the authors did not document the hepatitis viral load levels of the included patients, and both HBV and HCV can contribute to HCC to varying degrees. Additionally, a recent clinical trial found that diabetes is a potential risk factor for cancer in patients with AAN (Su et al. 2024). Therefore, the increased HCC risk observed in these results may be confounded by the severity of the underlying diseases. In the same year, a study quantified levels of AL-I-dA adducts—a direct biomarker of AAI exposure—in liver tumor tissues from 209 patients who underwent surgical resection at Guilin Medical University Affiliated Hospital from February 2001 to April 2007 (Wang et al. 2022). Liver samples from AAI-treated and untreated mice were used as positive and negative controls, respectively. The investigation identified AL-I-dA adducts in 52.6% (110/209) of patients, with these individuals demonstrating significantly poorer clinical prognoses. However, this study exhibited notable limitations: its single-institution design and the notably high positivity rate raise questions about potential regional environmental contamination or unidentified collective AA exposure pathways. While the analysis accounted for HBV as a confounding factor, it overlooked the multifactorial etiology of hepatocellular carcinogenesis, such as contributions from HCV, metabolic disorders, or other environmental carcinogens. To strengthen causality, inclusion of a dose-dependent relationship analysis between clinical outcomes and quantified AA exposure levels would have enhanced the validity of the conclusions. Furthermore, expanding the cohort to diverse geographical regions and incorporating longitudinal exposure data could mitigate selection bias and better elucidate AA's role in hepatocarcinogenesis. A recent retrospective clinical study, which followed 337 patients diagnosed with AAN from January 1997 to December 2014, revealed no HCC cases during the follow-up period (Su et al. 2024). Urothelial carcinoma remained the predominant malignancy observed, with no correlation identified between AA exposure dosage and urinary tract cancer incidence. Thus, these findings do not substantiate a causal relationship between AA exposure and HCC. However, this study carries methodological limitations, notably the absence of a control group comprising cancer patients without AA exposure, which may introduce inherent selection bias. To address these gaps, the authors are currently conducting follow-up investigations to refine methodological rigor and further substantiate their findings. Importantly, the multifactorial nature of hepatocarcinogenesis—including variables, such as coexisting viral hepatitis, metabolic disorders, or environmental

carcinogens—was not comprehensively addressed, necessitating broader epidemiological validation.

Animal models

Given the absence of direct evidence from both the aforementioned HCC tissue sequencing data and clinical trial results to substantiate a causal relationship between AA exposure and hepatic carcinogenesis, which necessitates further validation through animal experimentation, a comprehensive study was conducted utilizing AA-exposed murine models to elucidate the direct etiological connection. The investigation demonstrated that AAI exposure directly induced hepatocarcinogenesis in 2-week-old juvenile mice through a dose-dependent mechanism (Lu et al. 2020). Intriguingly, when combined with CCL4, a hepatotoxic agent, this co-exposure paradigm synergistically exacerbated hepatocellular carcinoma incidence, revealing a compound effect of dual hepatotoxic insults. However, due to the exclusive utilization of juvenile mice (developmentally equivalent to 2-year-old humans) in the original experimental design, this investigation carried inherent methodological constraints.

To address these limitations, a subsequent mechanistic study employing mature murine models demonstrated that chronic AAI administration failed to induce hepatocarcinogenic progression (Chen et al. 2022). This observation stands in diametric opposition to the protumorigenic effects previously documented in juvenile specimens, thereby highlighting critical age-dependent discrepancies in AA-associated hepatocarcinogenesis. The observed discrepancies may stem from several plausible mechanistic explanations: First, the route of administration diverged critically between studies. The juvenile murine model employed intraperitoneal injection, whereas the adult cohort received AAI via oral gavage—a clinically relevant route that better recapitulates human exposure patterns through decoction-based herbal ingestion. Second, distinct pharmacokinetic pathways were engaged: intraperitoneally delivered AAI bypasses first-pass intestinal metabolism, directly accessing hepatic circulation via mesenteric vasculature. Juvenile mice exhibit ontogenic deficiencies in hepatic expression and catalytic competence of AA-detoxifying enzymes (particularly CYP450 isoforms and glutathione-S-transferases), resulting in impaired toxic metabolite clearance (Peng et al. 2012). This developmental immaturity culminates in accelerated AA bioactivation, toxic metabolite accumulation, and consequent hepatocyte necroapoptosis. In contrast, mature murine livers demonstrate enhanced metabolic resilience through robust enzymatic detoxification, effectively mitigating AA-induced genotoxic stress. Third, the genetic mutations caused by carcinogens can be cloned and amplified through cell proliferation (Cohen and Ellwein 1990). The liver cells of 2-week-old

Table 2 Experimental findings pertaining to the impact of AA exposure on liver outcomes within animal models

Experimental model	AA exposure details	Results
Male Fischer 344 rats weighing 140–150 g (Rossiello et al. 1993)	10 mg/kg b.w., i.p	When AA is used alone, it does not possess hepatocarcinogenic properties, but when used after partial hepatectomy (PH) surgery, it can induce the formation of liver nodules in rats
Ten-month-old male beagle canines (Jin et al. 2016)	Canines were given capsules with AAI 3 mg/kg/day for 10 days	Hepatic premalignant alterations appeared in canines after a 10-day AAI oral administration
Male Sprague–Dawley rats weighing 180–220 g (Wang et al. 2021)	Divided into a 20 mg/kg AA group, a 4 mg/kg AA group, and a 2 mg/kg AA group. Each group of rats was administered the corresponding dose of AA by gavage for 28 days	AA caused mitochondrial disruption in the rat liver, decreased organelles, such as endoplasmic reticulum and ribosome, and caused inflammatory cell infiltration and fibrosis tendency in the liver
Six-week-old male Sprague–Dawley rats (Liu et al. 2021)	Medium-term study: For the initiation assay, a single oral dose of AAI (20, 50, or 100 mg/kg) was administered. For the promotion assay, doses of 0.1, 1, or 10 mg/kg AAI were administered once a day, 5 days a week (Monday–Friday), for a total of 6 weeks. Long-term study: AAI (0, 0.1, 1, or 10 mg/kg) was administered for up to 52 weeks [once a day, 5 days a week (Monday–Friday)]	AAI promoted clonal expansion in the liver only at a high dose of 10 mg/kg in adult rats, but these lesions had not progressed to HCC even in the presence of AAI-DNA adducts at the end of the study or until death caused by cancers that developed in the forestomach, intestine, and/or kidney
Two-week-old male C57BL/6J mice (Lu et al. 2020)	Subjected C57BL/6J male mice to AAI by i.p. at a dose of 2.5/5 mg/kg	AA can directly induce young mouse liver cancers, including HCC and cHCC-ICC
Eight-week-old C57BL/6, C57BL/6-TgHBV, BALB/c and BALB/c-TgHBV adult mice model (Chen et al. 2022)	For mice with AAI administration alone, AAI was given consecutively every other week by gavage. The total duration of AAI administration was 8 months	Long-term administration of AAI barely increased liver tumorigenesis in adult mice, opposite from its tumor-inducing role when subjected to infant mice. AAI induced dose-dependent accumulation of AA-DNA adduct in target organs in adult mice, with the most detected in kidney instead of liver
Two-week-old and eight-week-old male C57BL/6J mice (Fang et al. 2022)	From Monday to Friday, the control, low-dose, and high-dose groups were intraperitoneally injected with 0, 2.5, and 5.0 mg/kg/day AAI in the neonatal mice group, and with 0, 2.2, and 4.4 mg/kg/day AAI in the adult mice group	Hepatic cysts were observed in AAI-treated neonatal mice. However, there were no abnormalities in any of the adult mice groups. When exposed to equivalent doses of AAI, adult mice and neonatal mice could suffer different degrees of liver injury during different exposure times. Acute liver injury was generated in adult mice compared with neonatal mice

young mice are highly proliferative, making them more susceptible to genetic mutations that can lead to liver cancer.

In summary, disparate research designs coupled with subsequent metabolic variations may precipitate the observation of divergent hepatic effects. Herein, we have consolidated experimental findings pertaining to the impact of AA exposure on liver outcomes within animal models (Table 2).

In experiments utilizing adult rodents, the majority of studies did not observe HCC formation upon the sole administration of AA. This phenomenon may be attributed to the dosage of AA employed in certain investigations, which was insufficient to induce hepatic necrosis and, consequently, failed to elicit compensatory hepatocyte proliferation—a requisite condition for the initiation of carcinogenesis by carcinogens in the adult liver. Conversely, the administration of AA following partial hepatectomy (PH) has been associated with the observation of hepatic tumors, suggesting that AA can indeed instigate hepatocarcinogenesis when hepatocyte proliferation is heightened. This may also elucidate the elevated incidence of HCC in patients co-infected with HBV/HCV upon AA exposure, as the hepatic injury inflicted by hepatitis viruses may lead to increased hepatocyte proliferation, with AA consumption potentially accelerating oncogenesis. Furthermore, varying degrees of hepatic injury have been consistently observed in both juvenile and adult rodents following AA ingestion, underscoring the unequivocal hepatotoxicity of AA.

Conclusion and perspectives

In summary, there remains a paucity of conclusive evidence directly establishing AA exposure as an immediate causative agent of hepatocellular carcinoma. This necessitates meticulously designed animal studies with enhanced experimental rigor to conduct in-depth investigations, particularly addressing critical parameters, such as AA administration modalities and dosage optimization.

Furthermore, while AA-associated mutational signatures in hepatocellular carcinoma (e.g., COSMIC Signature 22) demonstrate potential as biological markers, their clinical utility in early diagnosis and prognostic evaluation requires comprehensive validation. The establishment of correlations between mutational burden and clinical phenotypes through large-scale cohort studies, coupled with the development of non-invasive detection methods utilizing liquid biopsy technologies, could substantially enhance hepatocellular carcinoma risk prediction capabilities.

Current understanding of the regulatory networks governing AA-metabolizing enzymes (including CYP1A1/2 and SULT isoforms) and their interindividual expression variability remains fragmentary. Integrating single-cell sequencing with epigenetic profiling could elucidate the genetic/

epigenetic mechanisms underlying metabolic enzyme activity, thereby clarifying the biological basis of population susceptibility and identifying targets for precision prevention strategies.

The frequent co-occurrence of AA exposure with environmental carcinogens, such as hepatitis viral infections and aflatoxin contamination, presents a complex carcinogenic milieu, yet their synergistic mechanisms remain enigmatic. Employing organoid-based models or multi-omics integrative analysis to delineate AA's interactions with other carcinogens in driving DNA damage, immune microenvironment remodeling, and epigenetic dysregulation will be crucial for validating the “second hit” hypothesis in hepatocarcinogenesis.

Exploration of therapeutic interventions targeting AA-DNA adduct repair mechanisms (e.g., base excision repair pathways) or mutation-associated signaling networks (particularly P53 regulatory cascades) through small-molecule inhibitors or gene-editing tools may pioneer novel avenues for chemoprevention and targeted therapy.

Moreover, strengthening global epidemiological surveillance of AA exposure and implementing stringent regulation of AA-containing phytochemicals in herbal medicines are imperative for mitigating hepatocellular carcinoma burden. Future research should adopt a multi-disciplinary framework integrating fundamental, clinical, and public health perspectives to construct a comprehensive scientific paradigm spanning mechanistic exploration to preventive intervention.

Acknowledgements Thanks to all the developers of the drawing software FigDraw.

Author contributions The majority content of this manuscript was written by YL. All authors made contributions to this manuscript. All authors have given approval to the final version of the manuscript.

Funding This study was funded by the Science and Technology Development Fund, Macau SAR (File no. 0075/2024/RIB2, 0105/2022/A2, and 006/2023/SKL).

Data availability Not applicable to this article as no new data were created or analyzed in this study.

Declarations

Conflict of interest The authors declare no competing financial interest.

References

- Alexandrov LB, Kim J, Haradhvala NJ et al (2020) The repertoire of mutational signatures in human cancer. *Nature* 578(7793):94–101
- Chen CH, Dickman KG, Moriya M et al (2012) Aristolochic acid-associated urothelial cancer in Taiwan. *Proc Natl Acad Sci USA* 109(21):8241–8246

- Chen CJ, Yang YH, Lin MH et al (2018) Herbal medicine containing aristolochic acid and the risk of hepatocellular carcinoma in patients with hepatitis B virus infection. *Int J Cancer* 143(7):1578–1587
- Chen CJ, Yang YH, Lin MH et al (2019) Herbal medicine containing aristolochic acid and the risk of primary liver cancer in patients with hepatitis C virus infection. *Cancer Epidemiol Biomarkers Prev* 28(11):1876–1883
- Chen S, Dong Y, Qi X et al (2022) Aristolochic acids exposure was not the main cause of liver tumorigenesis in adulthood. *Acta Pharm Sin B* 12(5):2252–2267
- Chen CJ, Chiu WC, Tseng YH et al (2022) Aristolochic acid and the risk of cancers in patients with type 2 diabetes: Nationwide population-based cohort study. *Phytomedicine* 99:154023
- Chen L, Zhang C, Xue R et al (2024) Deep whole-genome analysis of 494 hepatocellular carcinomas. *Nature* 627(8004):586–593
- Cohen SM, Ellwein LB (1990) Cell proliferation in carcinogenesis. *Science* 249(4972):1007–1011
- Connor F, Rayner TF, Aitken SJ et al (2018) Mutational landscape of a chemically induced mouse model of liver cancer. *J Hepatol* 69(4):840–850
- Cosyns JP, Jadoul M, Squifflet JP et al (1994) Chinese herbs nephropathy: a clue to Balkan endemic nephropathy? *Kidney Int* 45(6):1680–1688
- Danilovic V, Djuricic M, Mokranjac M et al (1957) Chronic nephritis due to lead poisoning by digestive route (flour). *Presse Méd* 65:2039–2040
- Dong H, Suzuki N, Torres MC et al (2006) Quantitative determination of aristolochic acid-derived DNA adducts in rats using 32P-postlabeling/polyacrylamide gel electrophoresis analysis. *Drug Metab Dispos* 34(7):1122–1127
- Dong YP, Chen SZ, He HS et al (2023) Skullcapflavone II, a novel NQO1 inhibitor, alleviates aristolochic acid I-induced liver and kidney injury in mice. *Acta Pharmacol Sin* 44(7):1429–1441
- Dubaisi S, Caruso JA, Gaedigk R et al (2019) Developmental expression of the cytosolic sulfotransferases in human liver. *Drug Metab Dispos* 47(6):592–600
- Fang ZE, Wang C, Niu M et al (2022) Integration of transcriptomic and metabolomic data to compare the hepatotoxicity of neonatal and adult mice exposed to aristolochic acid I. *Front Genet* 25(13):840961
- Ganshirt H (1953) Isolation of aristolochic acid from various Aristolochiaceae and its quantitative determination. *Pharmazie* 8(7):584–592
- Garaycochea JI, Crossan GP, Langevin F et al (2018) Alcohol and endogenous aldehydes damage chromosomes and mutate stem cells. *Nature* 553(7687):171–177
- Gillerot G, Jadoul M, Arlt VM et al (2001) Aristolochic acid nephropathy in a Chinese patient: time to abandon the term “Chinese herbs nephropathy”? *Am J Kidney Dis* 38(5):E26
- Gillet LC, Schärer OD (2006) Molecular mechanisms of mammalian global genome nucleotide excision repair. *Chem Rev* 106(2):253–276
- Gold LS, Slone TH (2003) Aristolochic acid, an herbal carcinogen, sold on the Web after FDA alert. *N Engl J Med* 349(16):1576–1577
- Hagemann U, Grase R, Thiele A et al (1982) Problems of drug safety: aristolochic acid. *MMW Munch Med Wochenschr* 124(25):611–612
- Hanawalt PC, Spivak G (2008) Transcription-coupled DNA repair: two decades of progress and surprises. *Nat Rev Mol Cell Biol* 9(12):958–970
- Heinrich M, Chan J, Wanke S et al (2009) Local uses of Aristolochia species and content of nephrotoxic aristolochic acid I and 2—a global assessment based on bibliographic sources. *J Ethnopharmacol* 125(1):108–144
- Hollstein M, Marion MJ, Lehman T et al (1994) p53 mutations at A: T base pairs in angiosarcomas of vinyl chloride-exposed factory workers. *Carcinogenesis* 15(1):1–3
- Huang J, Deng Q, Wang Q et al (2012) Exome sequencing of hepatitis B virus-associated hepatocellular carcinoma. *Nat Genet* 44(10):1117–1121
- Huang MN, Yu W, Teoh WW et al (2017) Genome-scale mutational signatures of aflatoxin in cells, mice, and human tumors. *Genome Res* 27(9):1475–1486
- IARC Working Group on the Evaluation of Carcinogenic Risks to Humans, International Agency for Research on Cancer, World Health Organization. Some traditional herbal medicines, some mycotoxins, naphthalene and styrene. World Health Organization, 2002.
- Ivić M, Lovrić B (1967) Carcinogenic action of Aristolochia. *Acta Medica Med.* 5:1–3
- Ji X, Feng G, Chen G et al (2018) Lack of correlation between aristolochic acid exposure and hepatocellular carcinoma. *Sci China Life Sci* 61(6):727–728
- Jin K, Su KK, Li T et al (2016) Hepatic premalignant alterations triggered by human nephrotoxin aristolochic acid I in canines. *Cancer Prev Res (Phila)* 9(4):324–334
- Kathuria P, Sharma P, Wetmore SD (2015) Adenine versus guanine DNA adducts of aristolochic acids: role of the carcinogen-purine linkage in the differential global genomic repair propensity. *Nucleic Acids Res* 43(15):7388–7397
- Levová K, Moserová M, Kotrbová V et al (2011) Role of cytochromes P450 1A1/2 in detoxication and activation of carcinogenic aristolochic acid I: studies with the hepatic NADPH:cytochrome P450 reductase null (HRN) mouse model. *Toxicol Sci* 121(1):43–56
- Liu Y, Chan CK, Jin L et al (2019) Quantitation of DNA adducts in target and nontarget organs of aristolochic acid I-exposed rats: correlating DNA adduct levels with organotropic activities. *Chem Res Toxicol* 32(3):397–399
- Liu YZ, Lu HL, Qi XM et al (2021) Aristolochic acid I promoted clonal expansion but did not induce hepatocellular carcinoma in adult rats. *Acta Pharmacol Sin* 42(12):2094–2105
- Llovet JM, Pinyol R, Yarchoan M et al (2024) Adjuvant and neoadjuvant immunotherapies in hepatocellular carcinoma. *Nat Rev Clin Oncol* 21(4):294–311
- Lord GM, Hollstein M, Arlt VM et al (2004) DNA adducts and p53 mutations in a patient with aristolochic acid-associated nephropathy. *Am J Kidney Dis* 43(4):e11–e17
- Lu ZN, Luo Q, Zhao LN et al (2020) The mutational features of aristolochic acid-induced mouse and human liver cancers. *Hepatology* 71(3):929–942
- Ma H, Wood TG, Ammenheuser MM et al (2000) Molecular analysis of hprt mutant lymphocytes from 1, 3-butadiene-exposed workers. *Environ Mol Mutagen* 36(1):59–71
- Magliulo E, Rondanelli R, Carcò FP et al (1966) Effects of aristolochic acid on phagocytosis. Research “in vivo” in humans with the Rebuck skin window technic. *Riv Emoter Immunoematol.* 13(5):113–28
- Manjanatha MG, Li EE, Fu PP et al (1996) H- and K-ras mutational profiles in chemically induced liver tumors from B6C3F1 and CD-1 mice. *J Toxicol Environ Health* 47(2):195–208
- Martínek V, Bárta F, Hodek P et al (2017) Comparison of the oxidation of carcinogenic aristolochic acid I and II by microsomal cytochromes P450 in vitro: experimental and theoretical approaches. *Monatsh Chem* 148(11):1971–1981
- Moese JR (1963) Alteration, of infection resistance by aristolochic acid. *Langenbecks Arch Klin Chir Ver Dtsch Z Chir* 5(304):657–660
- Moriya M, Slade N, Brdar B et al (2011) TP53 Mutational signature for aristolochic acid: an environmental carcinogen. *Int J Cancer* 129(6):1532–1536

- Nassar D, Latil M, Boeckx B et al (2015) Genomic landscape of carcinogen-induced and genetically induced mouse skin squamous cell carcinoma. *Nat Med* 21(8):946–954
- Nedelko T, Arlt VM, Phillips DH et al (2009) TP53 mutation signature supports involvement of aristolochic acid in the aetiology of endemic nephropathy-associated tumours. *Int J Cancer* 124(4):987–990
- Ng AWT, Poon SL, Huang MN et al (2017) Aristolochic acids and their derivatives are widely implicated in liver cancers in Taiwan and throughout Asia. *Sci Transl Med*. 9(412):eaan6446
- Peng L, Yoo B, Gunewardena SS et al (2012) RNA sequencing reveals dynamic changes of mRNA abundance of cytochromes P450 and their alternative transcripts during mouse liver development. *Drug Metab Dispos* 40(6):1198–1209
- Penning TM, Su AL, El-Bayoumy K (2022) Nitroreduction: a critical metabolic pathway for drugs, environmental pollutants, and explosives. *Chem Res Toxicol* 35(10):1747–1765
- Poon SL, Pang ST, McPherson JR et al (2013) Genome-wide mutational signatures of aristolochic acid and its application as a screening tool. *Sci Transl Med*. 5(197):197ra101
- Priestap HA, Torres MC, Rieger RA et al (2012) Aristolochic acid I metabolism in the isolated perfused rat kidney. *Chem Res Toxicol* 25(1):130–139
- Rosenquist TA, Einolf HJ, Dickman KG et al (2010) Cytochrome P450 1A2 detoxicates aristolochic acid in the mouse. *Drug Metab Dispos* 38(5):761–768
- Rossiello MR, Laconi E, Rao PM et al (1993) Induction of hepatic nodules in the rat by aristolochic acid. *Cancer Lett* 71(1–3):83–87
- Scarborough J (2011) Ancient medicinal use of Aristolochia: birthwort's tradition and toxicity. *Pharm Hist* 53(1):3–21
- Sidorenko VS, Yeo JE, Bonala RR et al (2012) Lack of recognition by global-genome nucleotide excision repair accounts for the high mutagenicity and persistence of aristolactam-DNA adducts. *Nucleic Acids Res* 40(6):2494–2505
- Songhan Wu (1964) Two cases of acute renal failure caused by mutong. *Jiangsu Zhong Yi* 10(2):12–13
- Stiborová M, Arlt VM, Schmeiser HH (2016) Balkan endemic nephropathy: an update on its aetiology. *Arch Toxicol* 90(11):2595–2615
- Su T, Fang ZE, Guo YM et al (2024) No incidence of liver cancer was observed in a retrospective study of patients with aristolochic acid nephropathy. *Chin J Integr Med* 30(2):99–106
- Sugasawa K, Ng JM, Masutani C et al (1998) Xeroderma pigmentosum group C protein complex is the initiator of global genome nucleotide excision repair. *Mol Cell* 2(2):223–232
- Tanchev Y, Evstatiev ZV, Dorosiev G et al (1956) Study on nephritis in the region of Vratza, in Bulgarian. *Sav Med* 7:14–29
- Tomlinson T, Fernandes A, Grollman AP (2020) Aristolochia herbs and iatrogenic disease: the case of Portland's powders. *Yale J Biol Med* 93(2):355–363
- Tong S (2005) Mechanism of HBV genome variability and replication of HBV mutants. *J Clin Virol* 34(Suppl 1):S134–S138
- Vanherweghem JL, Depierreux M, Tielemans C et al (1993) Rapidly progressive interstitial renal fibrosis in young women: association with slimming regimen including Chinese herbs. *Lancet* 341(8842):387–391
- Wang P, Bennett RA, Povirk LF (1990) Melphalan-induced mutagenesis in an SV40-based shuttle vector: predominance of A.T→T.A transversions. *Cancer Res*. 50(23):7527–31
- Wang Y, Ma X, Zhou C et al (2021) Aristolochic acid induces mitochondrial apoptosis through oxidative stress in rats, leading to liver damage. *Toxicol Mech Methods* 31(8):609–618
- Wang L, Bai SH, Song SJ et al (2022) Exposure to aristolochic acid I is associated with poor prognosis of liver cancer patients. *Toxicol Res (Camb)* 11(1):255–260
- Zhang J, Chan CK, Ham YH et al (2020) Identifying cysteine, N-acetylcysteine, and glutathione conjugates as novel metabolites of aristolochic acid I: emergence of a new detoxification pathway. *Chem Res Toxicol* 33(6):1374–1381
- 国家药品监督管理局.关于取消关木通药用标准的通知 [EB/OL]. (2003–04–01). <https://www.nmpa.gov.cn/xxgk/fgwj/gzwj/gzwjyp/20030401010101717.html>. Accessed 17 Mar 2025
- 国家食品药品监督管理局.关于加强广防己等6种药材及其制剂监督管理的通知[EB/OL]. (2004–08–05). <https://www.nmpa.gov.cn/xxgk/fgwj/gzwj/gzwjyp/20040805010101598.html>. Accessed 17 Mar 2025
- 戴忠,刘静,郭日新等. 含马兜铃酸的中药现状及相关问题思考[J]. 中国现代中药. 2024,26(08):1283–1293

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.